Linear Mixed Models

Ciaran Evans

Warmup

Work on the warmup activity (handout).

Warmup

Data on 37 undergraduate music majors. Each row represents a performance, and each musician has between 2 and 15 performances in the data.

- `id`: a unique identifier for the musician
- `na`: negative affect score (a measure of anxiety)
- `large`: whether the musician was performing as part of a large ensemble (large = 1), or as part of a small ensemble or solo (large = 0)

Question: What are the groups in the data, and what are the observations within each group?

groups: musicians

observations: each performance

Warmup

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Question: What mixed effects model should the researchers use to investigate their hypothesis?

$\text{Anxiety}_{ij} = \beta_0 + u_i + \beta_1 \text{large}_{ij} + \varepsilon_{ij}$

i = group (musician)

j = performance

$\beta_0 + u_i$: intercept for musician i (Expected anxiety level for a small ensemble/solo performance)

$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$

$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

Models with a random intercept

$$Y = X\beta + Zu + \varepsilon$$

$$u \sim N(0, \sigma_u^2 I_m)$$

$$\varepsilon \sim N(0, \sigma_\varepsilon^2 I_n)$$

$$Z = \begin{bmatrix} 1_{n_1} & 0 & 0 \cdots 0 \\ 0 & 1_{n_2} & 0 \cdots 0 \\ \vdots & & \ddots \\ 0 & 0 \cdots & 1_{n_m} \end{bmatrix}$$

1_{n_i} = vector of all 1s of length n_i

u, ε, X are all independent

$$Y \mid X \sim N(\mathbb{E}[X\beta + Zu + \varepsilon \mid X], \operatorname{Var}(X\beta + Zu + \varepsilon \mid X))$$

$$= N(X\beta, \operatorname{Var}(Zu) + \operatorname{Var}(\varepsilon))$$

$$= N(X\beta, Z(\sigma_u^2 I_m)Z^T + \sigma_\varepsilon^2 I_n)$$

$$= N(X\beta, \underbrace{\sigma_u^2 ZZ^T + \sigma_\varepsilon^2 I_n}_{\Sigma})$$

$$ZZ^T = \begin{bmatrix} 1 1^T & 0 & \\ 0 & 1 1^T & \\ & & \ddots \\ & & 1 1^T \end{bmatrix} = \begin{bmatrix} \cdots & & 0 \\ & \ddots & \\ 0 & & \cdots \end{bmatrix}$$

Models with a random intercept

$$\sigma_u^2 ZZ^T + \sigma_\varepsilon^2 I_n$$

$$\begin{bmatrix} \sigma_u^2+\sigma_\varepsilon^2 & \sigma_u^2 & \cdots & \sigma_u^2 & & & \\ \sigma_u^2 & \sigma_u^2+\sigma_\varepsilon^2 & \cdots & \sigma_u^2 & & \mathbf{0} & \\ \vdots & & \ddots & & & & \\ \sigma_u^2 & \cdots & & \sigma_u^2+\sigma_\varepsilon^2 & & & \\ & & & & \sigma_u^2+\sigma_\varepsilon^2 & \sigma_u^2 & \\ & \mathbf{0} & & & \sigma_u^2 & \ddots & \\ & & & & & & \sigma_u^2+\sigma_\varepsilon^2 \end{bmatrix}$$

(n_1+n_1 and n_2+n_2 mark the two blocks)

$$= \begin{bmatrix} \Sigma_1 & 0 & \cdots & 0 \\ 0 & \Sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \\ 0 & 0 & \cdots & \Sigma_m \end{bmatrix}$$

Models with a random intercept

$$Y \mid X \sim N(X\beta, \Sigma) \qquad \Sigma = \begin{bmatrix} \Sigma_1 & & 0 \\ & \ddots & \\ 0 & & \Sigma_m \end{bmatrix}$$

$$\Sigma_i \in \mathbb{R}^{n_i \times n_i} \qquad \Sigma_i = \begin{bmatrix} \sigma_u^2 + \sigma_\varepsilon^2 & & \sigma_u^2 \\ & \ddots & \\ \sigma_u^2 & & \sigma_u^2 + \sigma_\varepsilon^2 \end{bmatrix}$$

$$\mathrm{Var}(Y_{ij} \mid x_{ij}) = \sigma_u^2 + \sigma_\varepsilon^2$$

$$\mathrm{Cov}(Y_{ij}, Y_{ik} \mid x_{ij}, x_{ik}) = \sigma_u^2$$

$$\mathrm{Cor}(Y_{ij}, Y_{ik} \mid x_{ij}, x_{ik}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}$$

intra-class correlation

(correlation within groups!)

$$\mathrm{Cov}(Y_{ij}, Y_{lk} \mid x_{ij}, x_{lk}) = 0$$

$l \neq i \Rightarrow$ independent (MVN)

groups are independent

Intra-class correlation

$$Y_{ij} = u_i + \mathbf{x}_{ij}^T \beta + \varepsilon_{ij}$$

The **intra-class correlation coefficient** (ICC) is the correlation between two observations within the same group i:

$$\rho_{\text{group}} := Cor(Y_{ij}, Y_{ik} | \mathbf{x}_{ij}, \mathbf{x}_{ik}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \quad (j \neq k)$$

```
Random effects:
 Groups       Name        Variance Std.Dev.
 neighborhood (Intercept) 1048     32.37
 Residual                 6762     82.23
Number of obs: 1561, groups:  neighborhood, 43
```

$\hat{\rho}_{group} = \frac{1048}{6762 + 1048} = 0.134$

About 13.4% of variability in rental price is explained by variability between neighborhoods after accounting for overall satisfaction

Parameter estimation

$$y|X \sim N(X\beta, \Sigma)$$

weighted least squares:

$$\hat{\beta} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y$$

and since $y|X \sim N(X\beta, \Sigma)$ then

$$\hat{\beta}|X \sim N(\beta, (X^T \Sigma^{-1} X)^{-1})$$

But, Σ is unknown.

Estimating variance components

MLE for Σ : $y|X \sim N(X\beta, \Sigma)$

$\Rightarrow L(\beta, \Sigma | y, X) = f(y | X, \beta, \Sigma)$ (mvN pdf)

$$\propto |\Sigma|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(y-X\beta)^T \Sigma^{-1}(y-X\beta)\right\}$$

maximize!

<u>However</u> : ML estimates of variance tend to be biased

Restricted maximum likelihood (REML)

Problem: ML estimates of variance components are biased, because we also have to estimate β

REML solution: Write down a new likelihood function for a transformation of **y** that does *not* depend on β

Idea: $y|x \sim N(X\beta, \Sigma)$

$(y - X\beta)|X \sim N(0, \Sigma)$ ← no β!

but requires us to know β?

Idea: if $a \in \mathbb{R}^n$ st $a^T X = 0$

then $a^T X\beta = 0$

$\Rightarrow a^T(y - X\beta) = a^T y$

$\Rightarrow a^T y$ does not depend on β!

Then $a^T y$ is an <u>error contrast</u>

How do we find vectors a st $a^T X = 0$?

REML

$$P = X(X^T X)^{-} X^T$$

$$PX = X \quad \Rightarrow \quad (I - P)X = 0$$

$\Rightarrow$ locn of rows of $I - P$